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SUBJECT CODE NO: - RNP-8003
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A & B) Mech/ Civil / CSE / AI & ML / EEE (NEP) Sem - I&II
Examination June/July 2025
Engineering Physics

[Time: 3 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

SECTION A**Q1 Solve the following questions****20**

1. In X-ray emission tubes, X-ray is emitted by the acceleration of
 - a) Atoms
 - b) Protons
 - c) Electrons
 - d) Neutrons
2. The celestial source of X-ray is
 - a) Stars
 - b) Quasars
 - c) Black Holes
 - d) Nebula
3. X-rays have larger wavelengths than which of the following?
 - a) Gamma rays
 - b) Beta rays
 - c) Microwave
 - d) Visible light
4. The critical angle is defined as the angle of incidence at which the angle of refraction is: _____
 - a) 0°
 - b) 90°
 - c) 180°
 - d) 45°
5. Which of the following lenses is used in magnifying glasses?
 - a) Convex lens
 - b) Concave lens
 - c) Cylindrical lens
 - d) Plano-convex lens
6. The free electron theory can explain the following property of metals:
 - a) Low thermal conductivity
 - b) High electrical conductivity
 - c) High elasticity
 - d) Magnetic properties of metals
7. In Newton's Rings, the distance between two consecutive dark rings:
 - a) Increases as we move towards the centre
 - b) Decreases as we move towards the centre
 - c) Remains constant
 - d) Depends on the wavelength of light
8. What is the region enclosed by the optical cavity called?
 - a) Optical Region
 - b) Optical System
 - c) Optical box
 - d) Optical Resonator

Q2	a) Explain the diameter of dark and bright rings.	05
	b) State and explain piezoelectric effect?	05
Q3	a) Explain cyclotron with suitable diagram.	05
	b) Explain the time dependent Schrodinger equation.	05
Q4	a) Explain the ruby Leaser	05
	b) What is superconductivity? Explain Meissner effect in superconductors?	05
Q5	a) Explain Davisson-Germer experiment.	05
	b) Explain G M counter.	05
Q6	a) State and explain spontaneous and stimulated emission.	05
	b) Write short notes Type -I and Type -II superconductor?	05
Q7	a) What is ultrasonic wave and state the properties	05
	b) Explain He- Ne Leaser with suitable diagram working	05